

User Testing Tasks

Sequential Transfer Learning for Medicinal Plant Identification: Usability and Interpretability Evaluation

Researcher: Hareeshwar Muthukumaran

Supervisor: Dr. Shamik Palit

Institution: University of Stirling

Date: November–December 2025

Overview

This document describes the testing protocol, participant tasks, and questionnaire used to evaluate the usability and interpretability of the medicinal plant identification model's interface and confidence score presentation.

Part 1: Testing Session Structure

Session Duration

Total time: 20–30 minutes

Location: Computer lab or quiet study room at University of Stirling

Session Flow

1. **Welcome and Briefing** (2 min): Researcher explains what the session involves
2. **Example Outputs Review** (5 min): Participant views 2–3 example model output screens
3. **Task Completion** (15–20 min): Participant completes 5 tasks
4. **Questionnaire** (3–5 min): Participant answers brief questionnaire
5. **Debrief and Thanks** (1 min): Researcher thanks participant

Part 2: Participant Tasks

Important Note: Participants are NOT being tested on plant knowledge or identification ability. The focus is entirely on **interface clarity and understandability**.

Task 1: Interface Clarity Rating

Scenario:

Researcher shows participant a sample model output screen displaying:

- A leaf image (medicinal plant)
- Predicted species name: "Tulsi (Holy Basil)"
- Confidence score: "89%"
- Optional: Category label "Traditional medicine use: Common"

Task Instruction:

"Please look at this screen. On a scale of 1 to 5, where 1 = very unclear and 5 = very clear, how easy is it to understand what the model is saying?"

Response: Participant indicates rating on paper form or verbally recorded by researcher.

Purpose: Measure overall interface clarity.

Task 2: Confidence Score Understanding

Scenario:

Researcher shows two example predictions:

- Prediction A: Species name with 92% confidence
- Prediction B: Species name with 61% confidence

Task Instruction:

"Looking at these two predictions, what do you think the confidence scores (92% and 61%) mean? In your own words, what do they tell you?"

Response: Participant explains their understanding verbally (researcher takes notes).

Follow-up Questions (if needed):

- "Does a high confidence score (92%) mean the prediction is definitely correct?"
- "What would a 50% confidence score mean to you?"

Purpose: Assess whether participants correctly understand that confidence scores indicate model certainty, not accuracy.

Task 3: Presentation Format Preference

Scenario:

Researcher shows two different ways of presenting the same result:

Format A (Text-based):

\\

Species: Ashwagandha (Winter Cherry)

Confidence: 87%

Status: Endangered species in Bangladesh

\\

Format B (Visual with icons):

\\

[Leaf Image] → Ashwagandha (Winter Cherry)

Confidence:  (87%)

Status:  Endangered

\\

Task Instruction:

"Which of these two formats is easier to understand? Why? Which would you prefer, and why?"

Response: Participant indicates preference (A or B) and provides reasoning.

Purpose: Gather feedback on optimal presentation formats for future interface design.

Task 4: Information Completeness

Scenario:

Researcher shows a model output and asks:

Task Instruction:

"If you were using this system to identify a plant you found, what additional information would be helpful? For example, would you like to know:

- Whether the plant is endangered?
- Geographic distribution?
- Medicinal uses?
- Other information?"

Response: Participant indicates preferences and explains reasoning.

Purpose: Identify missing information that would improve user understanding and decision-making.

Task 5: Usability and Accessibility

Scenario:

Researcher shows the complete model output interface and asks:

Task Instruction:

"Looking at the overall interface, please tell me:

1. Is it easy to find the information you need?
2. Are there any parts that are confusing or unclear?
3. Would you make any changes to improve it?"

Response: Participant provides qualitative feedback (researcher takes detailed notes).

Purpose: Identify usability issues and collect open-ended feedback.

Part 3: Participant Questionnaire

After completing all tasks, the participant completes a brief questionnaire:

Questionnaire Form

Participant Code: _____

Date: _____

Q1: Overall Clarity

On a scale of 1 to 5, how easy was it to understand the model's predictions and confidence scores?

- ☐ 1 = Very difficult
- ☐ 2 = Difficult
- ☐ 3 = Neutral/Average
- ☐ 4 = Easy
- ☐ 5 = Very easy

Q2: Confidence Score Understanding

Which statement best describes your understanding of the confidence score?

- ☐ It means the predicted plant species is definitely correct
- ☐ It means the model is uncertain about the prediction
- ☐ It indicates how confident the model is in its prediction (but not whether it's correct)
- ☐ I'm not sure what it means

Q3: Interface Design Preference

Which interface format did you prefer?

- ☐ Text-based (Format A)
- ☐ Visual with icons and charts (Format B)
- ☐ No preference
- ☐ Different format entirely

Q4: Usefulness for Decision-Making

If you were using this system to identify a medicinal plant for harvesting, how useful would the information provided be?

- ☐ 1 = Not useful
- ☐ 2 = Somewhat not useful
- ☐ 3 = Neutral
- ☐ 4 = Useful
- ☐ 5 = Very useful

Q5: Additional Comments (Optional)

What suggestions do you have to improve the system or interface?

Part 4: Researcher Observations

During the session, the researcher records:

- [] Time taken for participant to understand confidence score
- [] Any points of confusion or hesitation
- [] Spontaneous comments from participant
- [] Body language indicating frustration or clarity
- [] Technical difficulties or accessibility issues

Observation Notes:
